ADR4AI: LLM-Based Architectural Pattern Selection and ADR Generation for AI-Enabled Systems

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Working draft: Tasks 1.1–1.9 only. The expert-analysis protocol has been approved for this draft. The proposed governance procedures have been approved for this draft; the required permissions and safeguards remain pending confirmation. This document is not a complete paper or a submission-ready manuscript.

I. Methodology

A. Research Questions and Study Scope

We organize the study around three research questions (Table I). The computational protocol treats a user story and its acceptance criteria as a requirement case, with the system purpose providing context. It assesses each catalog pattern against each case independently. The assessment distinguishes applicability, non-applicability, and insufficient information; a positive classification requires a supporting user-story or acceptance-criterion reference. These classifications are model outputs, not independently established ground truth.

RQ1 characterizes the model's classification outcomes rather than measuring their correctness. RQ2 compares the patterns classified as applicable with a separate retrieval baseline. This baseline retrieves catalog candidates and reranks them to a final Top-5 without making an LLM-based applicability judgment. Overlap is therefore a descriptive relationship between two output sets, not evidence of either approach's precision, recall, or superiority.

ADR generation is a separate stage: each pair classified as applicable supplies the requirement case, selected pattern, and applicability evidence for one generated ADR. Generation completeness and stored provenance establish the availability and traceability of these records, not their architectural quality. RQ3 examines expert assessments of these selected pairs and their corresponding ADRs, rather than adding another automatic applicability classification.

The prepared questionnaire measures decision adequacy for the stated requirements (Q1), decision grounding (Q2), presentation of relevant consequences and trade-offs (Q3), perceived usefulness as a decision record (Q4), and pattern applicability (Q5). These are five separate expert-assessed dimensions; they are not a validated composite quality score. Optional comments provide context for the ratings. Neither implementation feasibility, cost savings, novelty, nor actual improvement in architectural practice is directly measured by these items.

Experts evaluate cases for a system they know, and only pairs selected by the LLM are presented. Consequently, RQ3 can characterize expert judgments of the presented recommendations but cannot identify missed applicable patterns or estimate recall over the full set of assessed pairs. Expert agreement with a recommendation does not by itself establish implementation success or objective correctness. Repeated ratings from an expert and multiple cases from a system must not be treated as independent replications when interpreting the results. Conclusions remain bounded to the sampled requirements, catalog, generated records, and participating experts.

B. Industrial Requirement Sample

The study uses a bounded industrial sample comprising 4 systems, 12 user stories (3 per system), and 35 acceptance criteria (Table II). The supplied data-scope document defines this as a minimal, delimited sample for studying architectural-pattern selection and decision documentation for AI-enabled systems. It is not a representative sample of industrial requirements or AI-enabled systems. The available artifacts do not establish a probabilistic sampling procedure; we therefore do not interpret these counts as population estimates.

Each requirement case consists of the system purpose, one stakeholder user story, and all of that story's acceptance criteria. The source CSV stores one acceptance criterion per row, repeating the corresponding purpose and story. Grouping by the stable user-story identifier reconstructs the case without rewriting the original Portuguese requirement text or changing criterion order. The criterion-row count must not be confused with the number of requirement cases.

System purpose provides contextual information, but is not sufficient on its own to establish pattern applicability. Positive evidence must be grounded in the user story and/or its acceptance criteria. The model is not authorized to infer unstated technologies, requirements, or constraints from the purpose.

The source uses generic names `System_01`–`System_04`; the paper uses the corresponding stable identifiers `SYS01`–`SYS04`. Story and criterion identifiers preserve traceability within each system without exposing real system names. The organization is not named in this review draft. According to the supplied scope document, the study input is restricted to purposes, identifiers, user stories, and acceptance criteria; it does not include operational databases, personal data, credentials,

TABLE I
RESEARCH QUESTIONS, UNITS OF ANALYSIS, AND EVIDENCE MAPPING.

ID	Research question	Unit, evidence, and intended analysis
RQ1	How does the LLM classify architectural-pattern applicability across the requirement cases, and which patterns does it identify as applicable?	Unit: one user-story–pattern pair. Evidence: the official exhaustive assessment log, original requirements, and pattern catalog. Analysis: distributions of the three classifications and applicable-pattern sets by story and system, retaining zero-positive stories.
RQ2	To what extent do the patterns classified as applicable by the LLM overlap with the frozen RAG baseline's final Top-5 candidates for the same requirement cases?	Unit: one LLM-positive user-story–pattern pair, grouped by story. Evidence: official applicability assessments and the frozen final Top-5 rankings. Analysis: membership-intersection counts by story and overall, with the LLM-positive set as an explicit denominator; no applicability-accuracy metric.
RQ3	How do experts assess the applicability of the LLM-selected patterns and the adequacy, grounding, consequences, and perceived usefulness of the corresponding generated ADRs?	Unit: one expert's rating of one selected pair and its ADR on one questionnaire item. Evidence: consolidated expert responses to Q1–Q5 and optional comments for the prepared cases. Analysis: item-specific ordinal distributions and case-level summaries with response counts; comments contextualize ratings. Aggregation and inclusion rules must be specified before consolidated analysis.

TABLE II
THE BOUNDED INDUSTRIAL REQUIREMENT SAMPLE.

System ID	User stories	Acceptance criteria
SYS01	3	6
SYS02	3	11
SYS03	3	7
SYS04	3	11
Total	12	35

customer records, or production content. This delimitation does not itself establish participant consent, ethics approval, or permission for unrestricted public release.

C. Architectural Pattern Catalog

We use 70 pattern records from the repository associated with Heiland et al.'s multivocal literature review of design patterns for AI-based systems [1], [2]. For provenance verification, we compared the local files with the public repository at revision `9ff3e61` (the full revision is recorded in the artifact audit). All 70 records matched the corresponding repository files byte-for-byte, as checked using Git blob hashes. This identifies the source snapshot of the catalog; it does not establish the effectiveness of the individual patterns.

Each pattern is stored as one JSON record with a stable three-digit identifier, from `001` through `070`. The applicability prompt supplies exactly five catalog fields: *ID*, identifying the record; *name*, naming the pattern; *motivation*, describing the problem or need; *solution*, describing its proposed response; and *consequences*, describing resulting effects and trade-offs. Identifiers remain strings, preserving leading zeros and enabling joins with the assessment and ADR artifacts. Source text is preserved rather than normalized or rewritten.

The records also contain aliases, examples, related-pattern identifiers, categories, and resource identifiers. These additional fields are not supplied to the applicability model. Resource identifiers provide source pointers within the repository, not independently verified bibliographic descriptions in the local catalog. We therefore attribute the collection to the verified repository and its associated publication rather than infer individual source citations from numeric resource codes.

All five input fields are present as strings in every record. Explicitly empty motivation or consequences strings are retained as empty inputs, without inventing replacement descriptions. A missing field or non-string value is an input-validation failure, not an invitation to repair the source. This preservation policy keeps the assessed catalog traceable while acknowledging uneven descriptive completeness.

The catalog includes general architectural patterns, such as Client–Server, alongside patterns concerning machine-learning safeguards and model serving. We use the records as a bounded candidate set, not as equally mature, mutually independent, or universally applicable solutions. Catalog inclusion alone does not establish applicability to a requirement case; that relationship is assessed separately.

D. Exhaustive Pattern-Applicability Assessment

The experiment crosses all 12 requirement cases with all 70 catalog patterns, yielding $12 \times 70 = 840$ planned pairwise assessments. Each request contains one complete requirement case and one pattern, with no prior assessment or conversation history supplied. This exhaustive design does not retrieve or prefilter candidates. Separate requests do not imply statistical independence: assessments share stories, systems, catalog entries, and a model. The resulting classifications are model predictions, not expert-provided ground-truth labels.

a) Model and structured response: The recorded model identifier is `gpt-6-sol`, with reasoning effort `high`. The implementation uses the Responses API with strict JSON-schema output named `pattern_assessment`. The schema requires exactly three fields: `classification` (one of the three labels below), `requirement_refs` (an array of strings), and `rationale` (a string); additional properties are disallowed. Each stored result also includes the story ID, pattern ID, model identifier, and reasoning effort. No temperature or seed is set in the request, and the recorded model identifier does not pin a dated model snapshot.

b) Decision and grounding rules: The instructions distinguish:

- `applicable`: the story and/or its acceptance criteria provide positive evidence of a need, problem, constraint, or quality concern addressed by the pattern.
- `not_applicable`: those requirements provide evidence that the pattern does not fit the stated need, problem, constraint, or context.
- `insufficient_information`: the story and criteria do not provide enough evidence to establish either applicability or non-applicability.

System purpose supplies context but cannot alone justify applicability. The prompt prohibits assuming unstated requirements, technologies, constraints, or system characteristics. Absence of contradiction, possible usefulness, or the possibility of implementing, testing, or improving a requirement using a pattern is not sufficient positive evidence. Assessment concerns underlying architectural intent rather than literal terminology. An applicable response must cite at least one identifier belonging to the current story or one of its criteria, and briefly explain the evidence–pattern relationship; it must not propose an architecture, implementation, technology, or new requirement. These are prompting and reference-validation constraints, not proof that a model's rationale is semantically correct.

c) Execution, retries, and validation: Execution is sequential, in CSV story order and ascending pattern-ID order. SDK retries are disabled. The runner permits at most 3 attempts per pair, revisiting unresolved pairs in up to 3 ordered passes rather than immediately retrying each failed request. Before a request, it atomically persists an attempt reservation, so interrupted requests consume the attempt budget. A request failure updates that entry with its error type and a whitespace-normalized message, truncated to 300 characters with the configured API key redacted. Successful validation removes the pair from this unresolved-failure state.

The response must have completed, contain no refusal, and provide parseable structured output. Local validation checks the exact fields, label, reference array and identifier membership, a nonempty reference array for `applicable`, and a string-valued rationale. These checks do not require a nonempty rationale string or test the factual entailment of cited text. Only validated results are appended to the official log, flushed, and synced to disk; storage failures stop execution rather than risk duplicate writes. A nonblocking file lock prevents concurrent official runners.

On resume, the runner validates existing JSONL records, model configuration, pair uniqueness, and failure-state attempt counts. Malformed records, unknown identifiers, and interrupted writes lacking a final newline stop the run rather than being repaired or discarded. Completed pairs are skipped; unresolved attempt counts are retained, and exhausted pairs are not retried. The final on-disk check requires exactly all 840 expected pairs and 70 records per story before exporting story-specific applicable-pattern selections.

d) Official dataset and replication artifact: The official dataset is the top-level `assessments.jsonl` under the pattern-selection experiment's `outputs` directory, accompanied by `failures.jsonl`. Separate `pilot_SYS01-US01` directories (including the `_v1` and `_v2` variants) and `test_assessment.json` and `test_assessment_v2.json` are excluded, not merged with official records. An offline audit verifies complete official coverage and records input/source hashes, the exact instructions, JSON schema, and input template. The template presents system purpose, identified story and criteria, followed by the five pattern fields in Section I-C; the full instructions remain in the artifact rather than this space-bounded description.

The frozen log contains 840 validated pair outcomes and an empty final unresolved-failure file. This establishes completion, not that no retries occurred: successful records omit attempt counts, and successful retries remove failure-state entries. Thus, 840 describes the planned first-pass requests and retained outcomes, not an independently verified total number of API attempts. We do not claim exact stochastic reproduction from the stored configuration.

E. Retrieval and Reranking Baseline

We compare the LLM-positive pairs with a separate retrieval/reranking baseline, called the RAG baseline in the artifacts. Despite that name, this pipeline performs no generative step: it uses no external LLM to classify applicability or generate ADRs. Its final rankings are candidate recommendations, not validated applicability judgments.

a) Query and catalog representation: The baseline groups the English requirement CSV by story ID, retaining the CSV encounter order. One query per story concatenates labeled system purpose, the user story preceded by its bracketed identifier, and all acceptance criteria, each preceded by its bracketed identifier. Identifiers are preserved in the query and output for traceability, not substituted for requirement text. An offline audit verifies that the story/criterion identifiers and their ownership match the original sample. The baseline uses the English requirement copy, whereas the applicability experiment uses the original Portuguese text; this is not a language-controlled comparison, and identifier alignment does not itself establish translation equivalence.

The retrieval corpus contains one document per catalog pattern. It includes the pattern ID and nonblank source values for name, aliases, motivation, solution, consequences, examples, categories, related patterns, and resources; list entries are joined with semicolons, omitting blank entries. These catalog records match the original pattern fields, but the retrieval representation is richer than the five-field applicability input. The Chroma collection is versioned by a fingerprint of catalog content, text-format version, and embedding-model identifier. Index compatibility checks verify complete IDs, document text, and metadata before reuse, rebuilding incompatible contents.

b) Vector candidates and final ranking: The embedding model is `BAAI/bge-small-en-v1.5`. Chroma vector similarity search retrieves 15 candidate documents per query

(vector Top-15), using the library's default similarity configuration without overriding the distance metric. The cross-encoder `BAAI/bge-reranker-base` scores each query–candidate pair. Candidates are sorted by descending reranker score, with vector rank breaking ties, to retain the final Top-5. The implementation requires exactly 15 vector candidates and one finite reranker score per candidate. Thus, Top-5 is a reranked subset of Top-15, not the first five vector results. The stored output preserves the vector ordering and the final reranked ordering separately.

c) Candidate-overlap measure: Let S be the set of sampled stories, A_s the set of pattern IDs classified as `applicable` by the LLM for story s, and R_s the set of IDs in the frozen final Top-5 for that same story. For each LLM-positive pair (s, p), define $m(s, p) = 1$ when $p \in R_s$, and $m(s, p) = 0$ otherwise. The per-story overlap count is $o_s = |A_s \cap R_s|$. The pooled LLM-positive-set overlap proportion is

$$O = \frac{\sum_{s \in S} |A_s \cap R_s|}{\sum_{s \in S} |A_s|}, \qquad \sum_{s \in S} |A_s| > 0. \tag{1}$$

This weights positive pairs, not stories equally; it is not an average of story-level percentages. A zero-positive story remains in the sample with $o_s = 0$, but contributes neither a positive pair nor a per-story percentage. If no story has a positive pair, O is undefined, not zero. Only final Top-5 membership enters this measure: vector Top-15 membership, ranking position, and reranker scores do not. Since A_s consists of model predictions rather than independently established truth, O measures candidate overlap, not precision, recall, accuracy, ground-truth applicability, or superiority of either approach. A nonmatch does not demonstrate a missed applicable pattern.

d) Frozen source and reproducibility limits: The comparison uses `rag_baseline.csv` in the RAG experiment's `outputs` directory, not the separate `rag_baseline_reproduced.csv`. An offline audit validates one frozen row per story, unique known IDs, ID/name alignment, and final Top-5 inclusion in vector Top-15, without recomputing rankings. The package records fixed model identifiers and pins direct dependencies and numerical runtimes, but does not pin downloaded model revisions or fully lock transitive dependencies. Rebuilt approximate vector indices and numerical hardware differences may change ranks. Catalog/text compatibility checks do not eliminate these sources of variation. No embedding models, reranker, or index were executed for this manuscript audit, and exact end-to-end reproduction is not established by these structural checks.

F. Architecture Decision Record Generation

Figure 1 summarizes the pipeline and its separate retrieval comparison branch. ADR generation consumes the 36 unique (`user_story_id`, `pattern_id`) pairs classified as `applicable`. It produces one proposed record per pair, not a jointly optimized architecture or an expert-approved decision. The accepted protocol pilot is included in the 36 records.

a) Inputs and prompting: Each request supplies the system purpose, identified story and all criteria; the selected pattern's ID, name, motivation, solution, and consequences; and the prior applicability assessment's evidence references, corresponding requirement text, and rationale. RAG rankings and overlap flags are excluded. The prompt permits legitimate empty catalog descriptions and limits generation to one contextualized architectural decision. It prohibits inventing unsupported requirements, technologies, components, protocols, or implementation details, or merely repeating that the system will adopt the named pattern. System purpose is context, while the story and criteria supply requirements. The proposed decision must instantiate the pattern's architectural intent; consequences must follow from the supplied information rather than portray the decision as universally beneficial.

The Nygard-inspired structure [3] comprises a short title, neutral context describing the relevant forces, an active-voice decision phrased as "We will ...", and relevant positive, negative, and neutral consequences. These are prompting rules, not automatically verified semantic properties. One synthetic recruitment/compliance example demonstrates the transformation using catalog Pattern `007`, which is outside the selected ADR-generation pairs. The example is instructional, not an empirical case. The generated ADR must not mention the model, experiment, applicability process, or evidence identifiers. Source references and rationale remain in the surrounding research artifacts for traceability.

b) Configuration and output validation: Generation uses `gpt-6-sol` with reasoning effort `high` and one independent Responses API request per ADR, without prior generated ADRs in its input. The client disables SDK retries and sets a 600-second timeout. No temperature or seed is supplied, and the model identifier is not a dated snapshot. Strict JSON-schema output, named `adr_pilot` for both pilot and full execution, requires exactly four string-valued fields: title, context, decision, and consequences, disallowing additional properties. Local validation rejects incomplete responses, refusals, duplicate JSON keys, incorrect field sets, and empty or non-string fields. The application then adds the fixed status `Proposed`; the model neither selects nor outputs that status. Neither schema compliance nor a Proposed status establishes architectural adequacy or stakeholder acceptance.

Each completed JSON artifact stores its pair identifiers, model and reasoning effort, response ID, `api_calls=1`, source-file path and SHA-256, rendered-prompt SHA-256, strict-output validation flags, and the five ADR fields. The source hash identifies the pre-ADR selected-pair CSV bytes, now retained in the backup: the current selected-pair CSV additionally contains an ADR column and therefore has a different file hash. The offline compiler validates pair coverage, metadata, source/prompt hashes, and fields before joining outputs by pair IDs. ADR text is retained as generated, not normalized, translated, or architecturally corrected.

c) Pilot, execution, and authorized retry: The research artifacts document approval of the `SYS01-US01`–Pattern `001` pilot. Full execution reuses that record without regenerating it

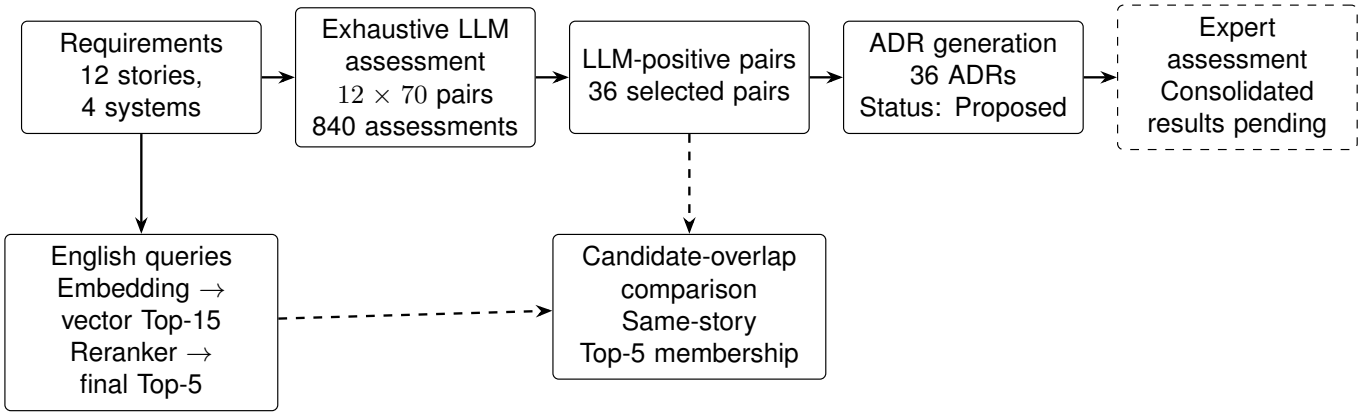


Fig. 1. ADR4AI pipeline. The retrieval branch supplies candidate rankings for comparison, not ADR-generation input. Dashed comparison arrows denote final-Top-5 membership; the dashed expert stage has no consolidated results in this draft.

and processes only pending pairs sequentially. It validates existing outputs and refuses duplicate or overwritten completed records. Persistent attempt claims prevent silent retries after interruptions; failure records retain the error, and any failure stops execution. Publishing full-run artifacts uses exclusive atomic creation rather than replacement.

For `SYS04-US03`–Pattern `003`, the archived failure reports "Connection error." A second attempt explicitly records manual retry authorization and links to the original claim. Both claims and the failure remain execution history, separate from the one completed ADR for that pair. Because a connection failure may precede dispatch, this failed attempt cannot be equated with an additional completed model response. The pilot plus final outputs cover exactly 36 unique selected pairs; attempt metadata is not counted as additional ADRs. Pilot approval concerned the generation protocol, not the future experts' judgments. Consolidated expert assessment is still pending in this draft.

G. Translation and Expert-Evaluation Preparation

a) Completed preparation: Portuguese presentation material: Evaluator-facing material was prepared for the 36 selected pairs. The original system purpose, story, and acceptance criteria are retained literally; translation operates on four pattern fields (name, motivation, solution, consequences) and five ADR fields (title, status, context, decision, consequences), yielding 324 field mappings. The linguistic inventory distinguishes 12 Portuguese fields preserved, 289 non-Portuguese fields translated, 14 mixed-language fields translated partially, and nine empty fields preserved. The translation prompt prohibits improving, completing, or architecturally correcting the source. Portuguese portions of mixed fields are protected by placeholders and restored locally; the fixed status `Proposed` is presented as *Proposto*. The preparation report records 36 translation requests in the prior translation stage; the subsequent review/compilation stage made no additional model calls.

Five field-specific linguistic corrections were explicitly authorized: "Commercial" to "Comercial" in four ADR fields for `SYS03-US02`–Pattern `014`, and "Machine Learning" to "aprendizado de máquina" in the pattern motivation for `SYS04-US03`–Pattern `014`. Original text, translation candidate, final presentation, action category, and correction authorization remain separate in the audit. An offline compiler checks exact source/mapping alignment, preserved Portuguese spans, authorized changes, requirement preservation, and pair equality before producing the presentation CSV. These checks establish linguistic provenance and preservation constraints, not architectural validation or independently measured translation quality.

b) Prepared instrument: items and routing: The archived Google Apps Script embeds all presentation cases verbatim. Three required profile questions ask about professional experience in software development/architecture/engineering, architecture-specific experience, and self-rated knowledge of the evaluated system. Both experience items use four categories: less than two years, two–five, six–ten, and more than ten; system knowledge uses a 1–5 scale from very low to very high. These items describe respondents but do not themselves enforce expert eligibility. The instructions ask evaluators to select only the system they know or have experience with; the required single-choice selection routes each evaluator to that system and then to the common final page, not onward to another system.

The prepared routes contain 7, 12, 8, and 9 cases for SYS01–SYS04, respectively. Within a route, purpose, story, and criteria are displayed once per story, followed by each pattern and ADR. Empty pattern descriptions are omitted from display without filling their stored values. Q1–Q4 address the ADR and Q5 its underlying pattern (Table III). All five ratings per case are required on the selected route, using a 1–5 agreement scale anchored by *Discordo totalmente* (strongly disagree) and *Concordo totalmente* (strongly agree); no verbal anchors are assigned to the intermediate values. Each case also has an optional paragraph comment asking about problems

TABLE III
PREPARED QUESTIONNAIRE CONSTRUCTS, IN PRESENTATION ORDER. ENGLISH DESCRIPTIONS OF THE PORTUGUESE ITEMS; NOT OBSERVED OUTCOMES.

Item	Judgment requested
Q1	Adequacy of the proposed decision for the story and acceptance criteria.
Q2	Sufficient grounding of the architectural decision to understand why it was proposed in that context.
Q3	Adequate presentation of relevant consequences, including benefits and possible impacts or trade-offs.
Q4	Usefulness, in the evaluator's system, as a record supporting understanding, discussion, or evolution of the decision.
Q5	Applicability of the underlying pattern to the story and acceptance criteria.

or needed improvements. Question titles encode the story ID, pattern ID, and item ID for traceable export. This yields 35, 60, 40, or 45 required case ratings per evaluator's route, not 180 ratings per evaluator. Unselected-route response columns are structurally blank, not evidence that the evaluator omitted selected-route ratings.

The five items are separate professional judgments, not a validated composite score. In particular, Q4 measures perceived usefulness as a record, not observed improvements in practice; Q5 cannot establish missed-pattern recall because only LLM-positive pairs are presented.

c) Implementation checks and unconfirmed administration: The questionnaire README reports local stand-in tests of item construction, embedded-case equality, and routing. Those simulations validate implementation, not expert judgments, Google deployment, live appearance, or account permissions. The manuscript audit independently checks frozen translation outputs, embedded values, and static questionnaire source contracts; it does not rerun the Google service or establish live routing behavior. Creation is disabled in the archived script until deliberately enabled by a researcher. The script requests no name, e-mail, or employee identifier, disables e-mail collection, response summaries, and question shuffling, and does not restrict submissions to one per signed-in user. These are intended settings, not a guarantee of anonymity: organization policies and responder permissions require account-side review and signed-out access testing. Profile combinations and free-text comments may also permit indirect identification.

d) Pending protocol and post-collection confirmation: **Protocol details pending confirmation:** document the deployed form/version and live route/access checks; evaluator eligibility and recruitment; consent and the applicable ethics approval or exemption; administration dates and instructions; and actual privacy, response-storage access, and retention arrangements. The preparatory research-purpose notice does not itself establish informed consent or ethics approval. The archived scripts alone do not establish that a form was deployed or that responses were collected.

The researcher reports preliminary responses being consolidated separately. No consolidated response dataset or verified administration record has been inspected for this draft. After collection, replace the pending procedural details with confirmed execution information, documenting any differences from the prepared instrument. This subsection contains no expert-result values or claims of agreement. Response eligibility, ordinal analysis, and comment handling are specified in Section I-H before response inspection.

H. Pre-Inspection Expert-Analysis Protocol

We record this exploratory, descriptive protocol before inspecting expert responses for manuscript analysis. Since the researcher reports preliminary responses already collected elsewhere, this is not a preregistration before collection. The versioned protocol records its timing, choices, output inventory, and future amendments; any change must disclose its reason and whether it followed exposure to results. Table IV predeclares the expert-result outputs. No response values are analyzed here.

a) Inclusion, routing, and missingness: Include submissions only after documented consent or applicable participation authorization, familiarity with the chosen system, and one recognized route with a compatible administered instrument are confirmed. Restricted administration records may supply confirmations absent from the form. No minimum years of experience or self-rated knowledge cutoff is imposed; profile questions alone do not prove eligibility. Unresolved eligibility is held pending, not inferred from favorable ratings. Retain an immutable raw-to-derived disposition log with included, excluded, and held-pending counts reconciling to the raw export.

Remove only proven duplicate export rows or submissions using provenance, not matching rating vectors or profiles. Hold unresolved duplicates and route conflicts pending investigation; do not silently select the more favorable response. Anonymous submissions are not necessarily unique people: report submission counts separately from confirmed distinct evaluator counts. Missing profile values do not remove otherwise eligible ratings.

Primary analysis retains available valid selected-route items from eligible submissions. Selected-route blanks and invalid values are recorded separately; neither is imputed. A valid rating is one integer category from 1 through 5, after whitespace trimming, without rounding or clipping. Blank cells on unselected routes are structural, outside eligible opportunities. Nonblank off-route ratings or ambiguous route selection quarantine the submission until export/instrument mapping is resolved. For each case/item, valid, missing, and invalid counts must sum to eligible opportunities. Repeat the analysis on complete-route submissions as a sensitivity check, reporting changed coverage even if no such submission remains.

b) Ordinal summaries and balanced aggregation: Analyze Q1–Q5 separately, without means, a composite score, significance tests, or treating ratings as independent participants. Report counts and percentages for all five categories among valid items, alongside opportunity, missing, invalid, and valid-response denominators. Use the non-interpolated inverse empirical distribution for the median and quartiles: the

TABLE IV
PREDECLARED EXPERT-ANALYSIS OUTPUT INVENTORY. THESE ARE PLANNED OUTPUTS, NOT OBSERVATIONS; AGGREGATE PUBLICATION REQUIRES DISCLOSURE REVIEW.

Output	Source and computation
Sample	Disposition log, profiles and administration confirmations. Reconcile row statuses; count route submissions, confirmed evaluators and profile categories with valid/unknown denominators.
Items	Eligible ratings and case-story-system mapping. System and balanced study distributions per item using the specified hierarchy; inverse-CDF median/quartiles, raw valid n and coverage.
Cases	Eligible ratings, cell flags and prepared case/item grid. Every case/item: opportunities, valid/missing/invalid n, category counts/percentages and quantiles; no-data summaries unavailable.
Hierarchy	Case/item distributions and prepared mapping. Equal-case story and equal-story system distributions, quantiles and observed/expected coverage; never average medians.
Figure	Primary item table's distributions. Stacked bars of all five ordered categories by system and balanced study summary, with coverage and identical scales.
Agreement	Confirmed-rater IDs joined to eligible ratings. Within-system/item ordinal alpha using the specified coincidence formula and gate; report coverage or non-estimability.
Sensitivity	Eligible complete-route submissions. Repeat case, hierarchy and item summaries unchanged; compare distributions and coverage with primary analysis.
Themes	Eligible comments, codebook, independent coding and reconciliation logs. Unique comments per theme with comment/opportunity denominators; preserve negative cases and multiple-code accounting.

smallest category whose cumulative proportion reaches 0.50, 0.25, or 0.75, respectively. For even sample sizes this yields the lower category median rather than an arithmetic midpoint. Display the median with its interquartile category interval $[Q_{0.25}, Q_{0.75}]$, without assuming equally spaced ratings. With no valid rating, summaries are unavailable, not zero.

First compute case/item distributions, giving each retained valid submission one vote within a case. Average observed case distributions equally within a story, story distributions equally within a system, and system distributions equally for the study summary. Apply the same quantile rule to these distributions, not to averaged medians. This avoids automatically overweighting systems with more recommended patterns. Report contributing and absent cases, stories, and systems; an observed-system summary with partial coverage must not imply all four systems were assessed. Raw counts and balanced percentages have different weighting and must be labeled. Retain item-wise within-submission summaries internally, without publishing respondent-level links to profiles. Stories with no selected pairs create no expert-rating opportunities and are not zero-score cases.

c) Conditional inter-rater agreement: Plan ordinal Krippendorff's alpha separately for each system and item, with ADR cases as units and confirmed distinct evaluators as raters. Do not pool raters assigned to different systems or equate unverified anonymous submissions with distinct raters. Compute alpha only when at least two confirmed raters share a case and expected disagreement is positive. The artifact specifies ordered-pair coincidence weighting, ordinal distance from empirical category marginals, and $\alpha = 1 - D_o/D_e$. Report rater and pairable-case counts, rating coverage, and any non-estimability reason; a single rater or invariant ratings do not justify an invented coefficient. Agreement is exploratory consistency, not correctness or agreement with the study hypothesis; no acceptance threshold is applied.

d) Comments and interpretation: Use each eligible nonblank case comment as the qualitative unit. Two researchers independently code the comments and reconcile differences, retaining initial assignments, codebook versions, and reconciliation reasons. Applicability, adequacy, grounding, consequences, usefulness, clarity, feasibility, missing context, and novelty are initial sensitizing domains, not guaranteed findings or additional questionnaire scales. Allow inductive subcodes and multiple codes, recoding affected comments after codebook changes. Preserve critical, mixed, and contradictory observations. Report unique comment counts per theme and comment opportunities; multiple themes may exceed the nonblank-comment total. Blank optional comments are not negative ratings. Quotations require documented authorization and de-identification; otherwise summarize without quoting. If dual coding cannot be executed, disclose the actual procedure and deviation.

There is no binary "expert-validated" criterion, favorable-rating cutoff, majority success rule, or post hoc threshold. Interpret the actual item-specific distributions and reservations with their coverage, distinguishing ADR judgments from Q5 applicability. They cannot establish missed-pattern recall, implementation success, or causal improvements in practice. All publication outputs require disclosure review; privacy decisions must not selectively suppress criticism. Consent/governance, instrument-version confirmation, and respondent provenance remain prerequisites to actual analysis. Phase 3 will implement and test the calculations against the frozen response export; this task records choices only.

I. Ethics, Data Governance, and Reproducibility

a) Authorization and confidentiality: The supplied industrial data-scope document defines the intended academic use of the bounded requirement sample and requires generic system identifiers. It is not evidence of formal organizational authorization, permission to transmit requirements to an external model provider, or permission for public redis-

tribution. **Confirmation pending:** document the responsible organization's authorization, permitted processing and publication scope, and any conditions on reuse. Generic identifiers conceal real system names but do not establish irreversible anonymization: requirement text and context may still identify the organization. Until disclosure is authorized, treat source requirements and requirement-bearing prompts, model outputs, translations, and evaluator cases as restricted; derived content does not automatically lose its source restrictions.

b) Expert participation and privacy: No formal ethics-board approval, exemption, or completed consent procedure is established by the supplied artifacts. **Confirmation pending:** record the applicable ethics determination, participant information and documented consent or participation authorization, recruitment and voluntariness safeguards, withdrawal conditions, actual platform access, authorized custodians, and retention/deletion arrangements. These conditions must be resolved before response analysis; a retrospective record must not be described as prior approval. The prepared form's settings do not prove anonymous administration. Raw exports, timestamps, profiles, comments, eligibility records, and any identity-linking keys require controlled access. Keep linking keys separate from analysis copies; generic submission IDs are pseudonyms, not proof of anonymity. Review small cells, combined profiles, and free text for deductive disclosure before publishing aggregates or authorized quotations, without selectively removing criticism.

c) External models and generated content: The archived applicability and ADR implementations use the OpenAI Responses API with the recorded `gpt-6-sol` identifier and high reasoning effort; the translation audit records model-assisted presentation preparation. The retrieval branch uses embedding and reranking models, not an LLM generation call. Requirement-bearing generation inputs and outputs must therefore be covered by organizational processing permission. **Confirmation pending:** verify the provider/account conditions in force at execution, including retention, training use, access, processing location, and any contractual safeguards; no zero-retention or non-training claim is inferred from source code. This manuscript-preparation step makes no model-generation calls. Generated classifications remain model judgments and ADRs remain `Proposed`; schema checks and translation corrections do not establish semantic correctness or deployment approval. Human review remains necessary before operational use.

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e) Sharing boundaries: The planned artifact package may contain disclosure-reviewed manuscript sources, processing code, analysis specifications, sanitized prompt templates, and permitted aggregate outputs. Public catalog provenance does not alone establish redistribution rights; verify source licenses and attribution. Requirements and dependent artifacts require explicit release permission and disclosure review. Raw expert data and identity-linking records remain restricted; approved de-identified derivatives are separate release candidates, not substitutes silently presented as raw evidence. The initial collection, including its identifying scope document, is not an anonymous review package. Review files, metadata, local paths, API response identifiers, and repository history before release. No public artifact URL or completed access-control implementation is claimed here; the availability statement will identify actual releases and restrictions once verified.

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